2025 IEEE Biomedical Circuits and Systems Conference (BioCAS) | 979-8-3315-7336-2/25/$31.00 ©2025 IEEE | DOI: 10.1109/BioCAS67066.2025.00013

Multimodal Signal Edge-Optimised Deep Learning Model for Real-Time Auditory Attention Classification

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***Abstract*—This work proposes a deep-learning framework for the real-time auditory attention classification using multimodal physiological signals, integrating electroencephalogram (EEG) and photoplethysmogram (PPG) signals. The architecture combines a multi-scale one-dimensional (1D) convolutional backbone to capture features across varying temporal resolutions, followed by a channel-wise attention mechanism using a Squeeze-and-Excite (SE) block to enhance feature selectivity. A Bi-directional Gated Recurrent Unit (Bi-GRU) is incorporated to model sequential dependencies within the multimodal time-series data. The trained model is quantised, optimised, and deployed on the STM32L496G-DISCO microcontroller using the X-CUBE-AI toolchain. Deployment on board demonstrated a low memory footprint (RAM: 57.91 KiB, Flash: 31.47 KiB) and real-time inference capability with an average latency of 476.9 ms and accuracy of 79.75%. The proposed model achieved an overall classification accuracy of 93.39%, Sensitivity of 88.72%, Precision of 91.69%, and F1-score of 92.07%, and AUC values of 0.96 across all classes.**

***Index Terms*—Embedded Deep Learning, Edge AI, Real-Time Inference, Low-Power Wearable Systems, Multimodal Learning.**

I. Introduction

The fusion of multimodal physiological signals has become a critical strategy for enhancing the robustness and accuracy of cognitive and affective state classification tasks, including auditory attention decoding, emotion recognition, and stress detection. Signals such as electroencephalogram (EEG) and photoplethysmogram (PPG) offer distinct but complementary insights—EEG captures cortical activity associated with auditory and attentional processing [1], while PPG reflects autonomic nervous system responses linked to arousal and emotional states [2], [3]. Multimodal frameworks that integrate these signals have shown superior performance over unimodal systems in applications such as emotion recognition [4]–[6], seizure prediction [7], and mental workload analysis [8], [9]. Despite the advancements in deep learning-based multimodal models, their deployment on edge devices, such as wearable health monitors and IoT-based systems, remains a significant challenge due to constraints in memory, processing power, and real-time responsiveness. Prior studies have highlighted the trade-off between model complexity and edge compatibility [10]–[12]. While several works have proposed lightweight architectures or used knowledge distillation techniques, achieving high inference accuracy within strict latency and resource budgets is still challenging [13].

In response to these challenges, this paper presents a novel deep learning framework that utilises multimodal signals. The proposed model is optimised for real-time execution and quantised for deployment on the STM32L496G-DISCO platform [13]–[15], demonstrating high classification accuracy and resource efficiency suitable for continuous health monitoring in edge-based wearable systems.

II. Proposed Methodology

This work introduces an energy-efficient deep learning framework for real-time state classification of the Physiology of Auditory Attention (PhyAAt) [16] database. The PhyAAT database is a multimodal physiological dataset used in this work to validate the performance of the methodology for activity detection. The attention scores of the database from the listeners are categorised into three classes: listeners with scores ≤ 25 are classified as not-active, those with scores below 70 and > 25 are categorised as moderately active, and those with scores between ≥ 70 and ≤ 100 are considered fully active. The detailed description of the proposed methodology is shown in Fig. 1. The methodology consists of the following essential stages: (a) Pre-processing, (b) Deep learning model architecture & validation, and (c) Edge Deployment and Op-

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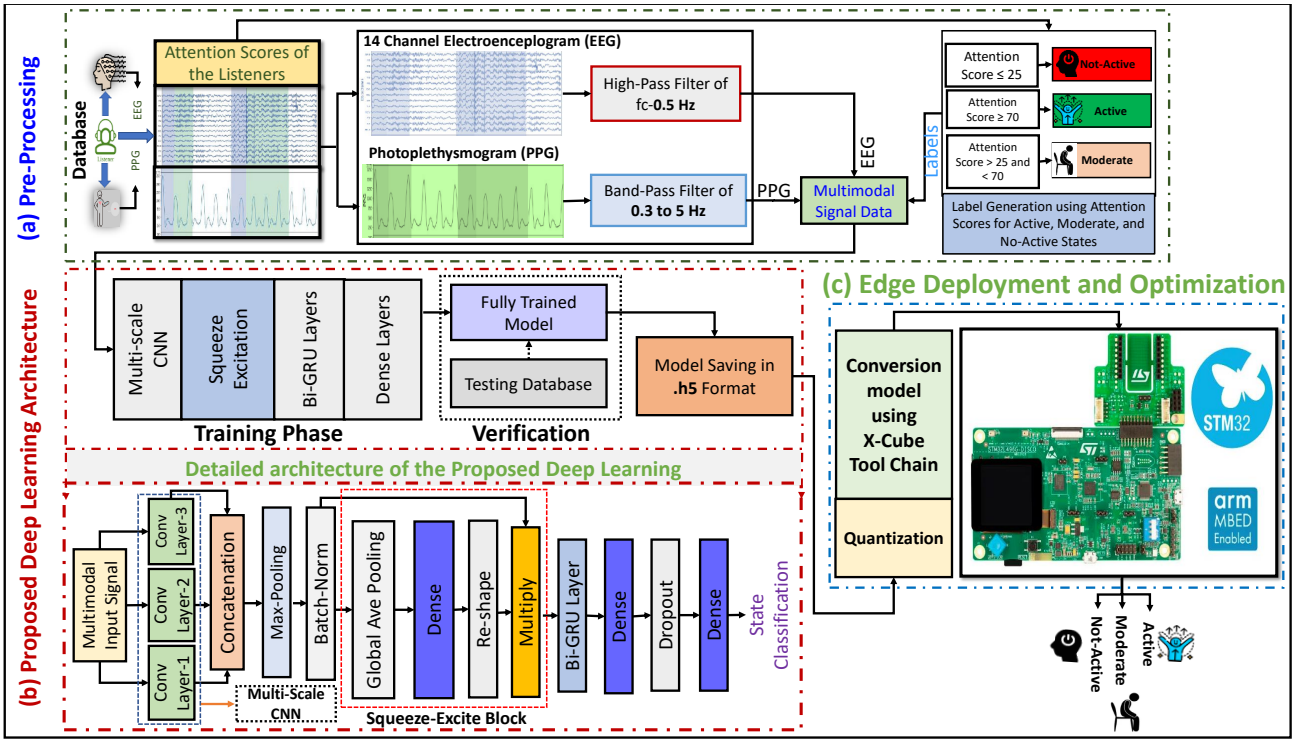


Fig. 1: An end-to-end proposed methodology for the multi-state classification.

TABLE I: Model Architecture Summary

Layer (type)	Output Shape	Param #
Input_layer	(1760, 1)	0
Conv1D	(1760, 64)	256
Conv1D_1	(1760, 64)	384
Conv1D_2	(1760, 64)	512
Concatenate	(1760, 192)	0
Max_pooling1D	(880, 192)	0
Batch_normalization	(880, 192)	768
Global_average_pooling1D	(192)	0
Dense	(24)	4,632
Dense_1	(192)	4,800
Reshape	(1, 192)	0
Multiply	(880, 192)	0
Bidirectional GRU	(128)	99,072
Dense_2	(128)	16,512
Dropout	(128)	0
Dense_3	(64)	8,256
Dropout_1	(64)	0
Dense_4	(3)	195

timisation. The details of these stages are discussed in the Subsections II-A, II-B, and II-C.

A. Pre-processing

To ensure signal fidelity, preprocessing is applied to both modalities. EEG signals are high-pass filtered (cut-off frequency f_c = 0.5 Hz, filter order n = 5) to eliminate low-frequency drifts and artifacts. The EEG spectrum is then decomposed into canonical frequency bands ranging from delta (δ) to high-gamma (γ_{high}), spanning 0.5–64 Hz [16]. Concurrently, PPG signals are processed using a band-pass Butterworth filter (0.3–5 Hz) to isolate physiologically relevant components [16].

B. Proposed Deep learning model architecture & validation

The proposed model integrates separate 1D convolutional layers with varying kernel sizes to capture features from modalities. A channel attention mechanism enhances the model's focus on informative features. The Squeeze-and-Excite (SE) Block recalibrates channel-wise feature responses, improving feature selectivity. Extracted features from multi-modal data are passed through fully connected layers with dropout regularisation to yield the final classification. Table I depicts the detailed model summary. The detailed procedure is also summarised in Algorithm 1.

C. Edge deployment and optimization

The trained model is first quantised to an 8-bit integer format to minimise memory consumption. It is then converted into *C* code using the X-CUBE-AI toolchain [17], enabling deployment on the microcontroller. Finally, on-device benchmarking is performed to evaluate inference latency, memory (RAM/Flash) usage, and classification accuracy.

III. Experimental Results&Discussion

This section evaluates the proposed framework for the real-time classification of physiological states. The dataset was

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Algorithm 1 Real-time Multimodal Auditory Signal for State Classification

Require: EEG signal $X_{\text{EEG}}(t)$, PPG signal $X_{\text{PPG}}(t)$
Ensure: Predicted state $\hat{y}_t$ at each time window t

Initialize sampling rate f_s, window size T, model parameters θ
for each window $t = 1$ to N **do**
  Preprocessing:
  Filter EEG: $X^f_{\text{EEG}}(t) \leftarrow \mathcal{B}(X_{\text{EEG}}(t))$
  Filter PPG: $X^f_{\text{PPG}}(t) \leftarrow \mathcal{A}(X_{\text{PPG}}(t), X^f_{\text{EEG}}(t))$
  Normalize: $\tilde{X}_{\text{EEG}}(t), \tilde{X}_{\text{PPG}}(t)$
  Fusion:
  $X_{\text{MM}}(t) \leftarrow \text{Concat}[\tilde{X}_{\text{EEG}}, \tilde{X}_{\text{PPG}}]$
  Feature Extraction:
  $F_c(t) \leftarrow \sum_k \text{Conv1D}_k(X_{\text{MM}}(t))$
  Channel Attention (SE):
  $w_c \leftarrow \sigma(W_2 \cdot \text{ReLU}(W_1 \cdot \text{GAP}(F_c)))$
  $F'_c(t) \leftarrow w_c \cdot F_c(t)$
  Temporal Modeling:
  $H(t) \leftarrow \text{BiGRU}(F'(t))$
  Classification:
  $\hat{y}_t \leftarrow \arg\max(\text{Softmax}(W_o \cdot H(t) + b_o))$
  if $\hat{y}_t \neq \hat{y}_{t-1}$ **then**
    Trigger alert or log decision
  end if
end for

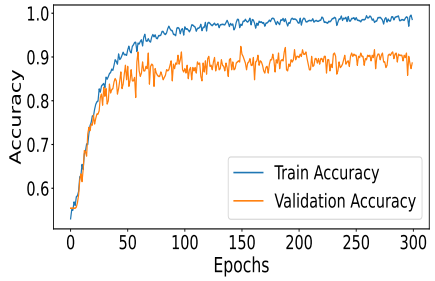
split into 80% for training and 20% for testing in a stratified manner to preserve class distribution. The initial experiments are conducted with EEG alone, and the accuracy of the EEG alone is 78.99%. Hence, we extended the experiments to multimodal data for better auditory-based state classification. The detailed metrics with the EEG and multimodal signal are shown in Table II.

TABLE II: Performance Metrics on the Testing Dataset

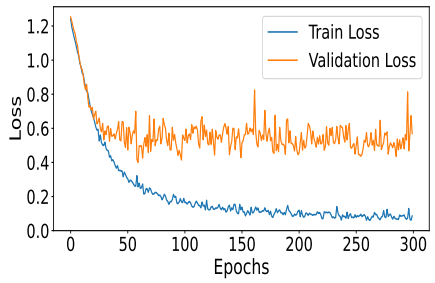
Class	Precision	Recall	F1-Score	Accuracy
EEG Only				
Not-Active	0.73	0.67	0.70	0.79
Active	0.83	0.87	0.85	0.79
Moderate	0.74	0.70	0.72	0.79
Multimodal (EEG + PPG)				
Not-Active	0.92	0.90	0.91	0.94
Active	0.92	0.97	0.94	0.93
Moderate	0.94	0.89	0.90	0.96

A. Training and Validation Performance

The training history of the proposed multimodal deep learning model is illustrated in Fig. 2(a) and Fig. 2(b), which shows the accuracy and loss curves over 300 training epochs. The model demonstrates stable convergence with consistent improvements in training and validation accuracy. Fig. 3 shows the confusion matrix of the predictions on the testing dataset, which shows the number of incorrect and correct predictions. Corresponding classification metrics are presented in Table II. To evaluate the discriminative performance of the



(a)



(b)

Fig. 2: Performance curves of the proposed method.

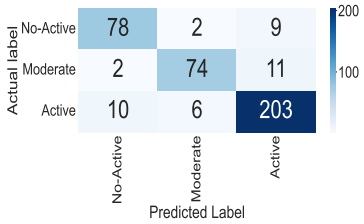


Fig. 3: Confusion matrix of the proposed method on the testing database.

proposed model, the Receiver Operating Characteristic (ROC) curves were plotted and are shown in Fig. 4. These results demonstrate a separation between the different physiological states' positive and negative class distributions.

B. Deployment on Edge Device

The MCU board was chosen as the deployment target and the model achieved an average inference time of approximately 476.9 ms across 16 forward passes, with a CPU workload of 47% (duty cycle of 1 second). The layer-wise latency analysis results in Table III validate the model's suitability for low-power embedded devices. Table IV summarises the layer-wise inference performance on the MCU. During edge deployment testing on the board, as shown in Table V, the model achieved a real-time classification accuracy of 79.75%, Root Mean Square Error of 0.3456, Mean Average Error of 0.2706, with precision of 76.48%, recall of 77.48%, and F1-score of 78.87%. While this represents a drop from offline

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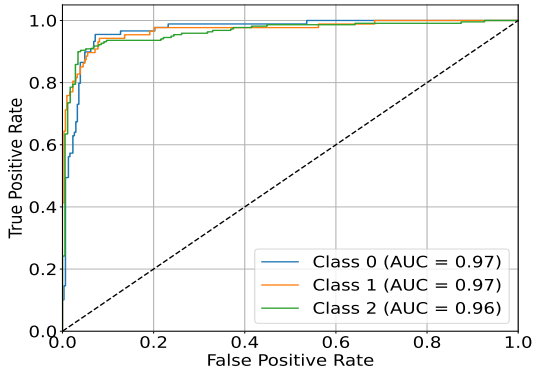


Fig. 4: ROC curve of the proposed method.

TABLE III: STM32L4 Deployment Summary

Metric	Value
Model Format	Float32 (Keras)
Input Shape	1×1760×1
Output Shape	1×3
Total Parameters	2,019 (7.89 KiB)
Total MACC	1.52 Million
Flash Usage	31.47 KiB
Weights	7.89 KiB
Runtime Library	23.58 KiB
RAM Usage	57.91 KiB
Activations	55.38 KiB
Runtime RAM	2.54 KiB
Top Compute Layer	GRU (90.7% MACC)
Flash Utilization	74.9%
RAM Utilization	4.4%

performance, it reflects the model's baseline performance under strict hardware constraints, offering a valuable benchmark for real-time embedded deployment.

C. Comparison of the proposed method with state-of-the-art

Table VI presents a comparative analysis of the proposed method against recent multimodal physiological signal classification literature. It highlights the proposed method's performance and deployment suitability. The proposed model achieves a better accuracy of 93.39%, and this end-to-end deployment from training to real-time embedded execution using the X-CUBE-AI toolchain demonstrates practical feasibility in wearable health monitoring, an advancement beyond most state-of-the-art methods that focus only on offline evaluation or require power-hungry hardware.

TABLE IV: Layer-wise Inference Performance on STM32L496G (80 MHz)

c_id	Layer Type	Time (ms)	Percentage (%)
0	OPTIMIZED_CONV1D	115.421	24.20%
1	GRU (BiGRU)	361.408	75.78%
2	Dense (dense_2)	0.044	0.01%
3	Non-Linearity (NL)	0.006	0.00%
4	Dense (dense_3)	0.015	0.00%
5	Non-Linearity (NL)	0.013	0.00%
Total Inference Time		**476.909**	**100.00%**

TABLE V: Confusion Matrix of Test Database on STM Board

Actual \ Predicted	C0	C1	C2
C0	70	6	13
C1	18	47	22
C2	9	12	198

TABLE VI: Comparison of the Proposed Method with Existing Literature

Literature	Database	Metrics	Purpose	Remarks
[4]	WESAD, CASE	Acc: 86.7%	Emotion recognition	ECG + EDA; high accuracy but complex model
[12]	WESAD, WEMAC	F1: 99.6%, Acc: 99.6%	Fear detection	Higher time on Edge TPU
[18]	Custom EMG/EEG	Acc: 93.4%	Gesture control	Event-driven decoder, optimized for FPGA-based edge inference
[5]	Wearable dataset	Acc: 95%, F1: 95%	Emotion detection	Temporal fusion from wearable sensor data, mobile DL deployment
Proposed Model	PhyAAT	Acc: 93.39%, F1 > 90%, AUC > 0.96	State classification	Compact, high-accuracy model deployed on STM32 MCU using X-CUBE-AI

IV. Conclusion

This paper proposes a lightweight yet robust deep-learning framework for the real-time classification of multimodal physiological signals. The model integrates multi-scale 1D convolutional layers, a channel attention mechanism using Squeeze-and-Excitation blocks, and a Bidirectional GRU to capture spatial and temporal dependencies across signal modalities effectively. Experimental results demonstrated high classification accuracy, with class-wise F1-scores exceeding 90% and AUC values above 0.96 for all classes. Using the X-CUBE-AI framework, the trained model was quantised and successfully deployed on the microcontroller. Layer-wise profiling confirmed that the architecture is optimised for embedded environments with limited computational resources. Overall, the proposed approach enables accurate, low-latency, and resource-efficient physiological state classification suitable for continuous health monitoring in wearable devices.

Acknowledgement

This work is supported by the KU-UAEU Joint Research Program Award No. KU-UAEU-2023-004 has the UAEU grant code G00004680.

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